MIL-C-60940 (MU)
10 January 1968

MILITARY SPECIFICATION

CUP, LEAD ASSEMBLY FOR FUZES, PD, M551 AND M533

1. SCOPE

1.1 This specification covers one type of lead cup assembly designated as Lead Cup Assembly for use in Fuzes, PD, M551 and M533.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

JAN-G-155 - Graphite (For use in Ammunition)

MILITARY

MIL-R-398 - RDX
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-L-10287 - Green Lacquer
MIL-C-20470 - Calcium Resinate
MIL-P-45486 - Pellets, RDX
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8876218 - Lead Cup Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8876218 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49-CFR 171-178 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite, "49 CFR 171-178, (latest revision)".)

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3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specifications.

RDX	- MIL-R-398, Type B, Class C
Calcium Resinate	- MIL-C-20470, Type II, except that the material shall pass a number 100 United States Standard Sieve
Graphite	- JAN-G-155, Grade I or II

3.2 Lead cup assembly.- The lead cup assembly shall comply with all requirements specified on Drawing 8876218 and with all requirements specified in applicable specifications.

3.3 Cracks or laminations.- The lead cup assembly, including the surface of the RDX shall be free of all cracks or laminations prior to application of the green lacquer.

3.4 Moisture content.- The moisture content of the RDX composition at the time and site of loading shall not exceed 0.20 percent when determined as specified in 4.4.5.

3.5 Functioning.- The lead cup assembly shall be capable of high order initiation by an M55 detonator across a minimum air gap of 0.165 inches when tested in accordance with 4.4.7. High order detonation will be evidenced by a depression in the lead block approximately 5/16 inch in diameter and a minimum of .125 inches in depth.

3.6 Workmanship.- All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests.- The Contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract, the specification, or as requested by the Contracting Officer. (See 6.5)

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.- Not applicable.

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4.3 Inspection provisions.-

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of lead cup assemblies shall contain:

a. Metal parts or metal parts assemblies of one interfix lot number from one manufacturer.

b. RDX from not more than one batch or blend of batches.

4.3.2 Examination.- Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with

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MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Lead cup assembly (see dwg. 8876218).-

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Weight, min.	Balance	(01001)
102.	Length	Gage	(01002)
103.	Diameter, max.	Gage	(01003)
104.	Cracks or laminations (See 3.3)	Visual	(01004)
Minor:	AQL 0.65 Percent		
201.	Green lacquer seal missing	Visual	(01005)

4.3.3 Testing.-

4.3.3.1 Composition (See dwg. 8876218), Major defect, Code No. (02001).-A sample of approximately four (4) ounces of the unconsolidated material shall be selected for the performance of these analyses. Failure to comply with all of the specified requirements shall be cause for rejection of the lot.

4.3.3.2 Moisture content of RDX (See Paragraph (Para.) 3.4); Major defect, Code No. (03001).-At the time and place of loading, a sample of approximately twenty (20) grams of unconsolidated material shall be selected from each lot and subjected to this test. If any sample fails to comply with the applicable requirement, the lot shall be rejected. The test shall be conducted as specified in 4.4.5.

4.3.3.3 Process functioning test (See Para. 3.5), Major defect, Code No. (04001).-Twenty (20) loaded lead cup assemblies shall be selected from each four (4) hours production and subjected to this test. If any assembly fails to comply with the applicable requirement, the four (4) hours production represented by the sample shall be rejected.

4.3.3.4 Functioning (See Para. 3.5), Major defect, Code No. (05001).-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.10

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percent. The test shall be conducted as specified in 4.4.7 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.- Equipment Tabulation (ET) number 8876218 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.- The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Determination of Calcium Resinate or Stearic Acid Content.- Transfer an accurately weighed portion of approximately 5 grams (gm) of the specimen to a tared medium porosity, sintered glass crucible. Extract the sample with three separate 20 milliliter (ml.) portions of chloroform which has been previously saturated with RDX, maintaining each portion in contact with the residue for approximately five minutes before applying suction. Stir occasionally to break up any lumps which may persist and wash down the sides of the crucible with each 20 ml. portion of

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chloroform to insure complete removal of the calcium resinate or stearic acid. Aspirate the residue until the odor of chloroform is no longer detectable, place the crucible and contents in a drying oven at 100 degrees Centigrade (°C.) to 105°C. for one half hour, cool in a desiccator and weigh. Repeat the above extraction, drying and weighing procedure with individual 20 ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 gm. Calculate the loss in weight of the crucible plus contents to percent calcium resinate or stearic acid, as applicable, on a moisture-free basis as follows:

$$\text{Percent calcium resinate or stearic acid} = \frac{100A}{W}$$

where:

A = Loss in weight of the crucible plus contents.
W = Weight of specimen on a moisture-free basis.

4.4.2 Determination of Graphite Content of Binder-Lubricant (when applicable).- Extract the residue remaining after the calcium resinate or stearic acid has been removed with two 20 ml. portions of dimethylformamide, maintaining each portion in contact with the residue for approximately five minutes with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20 ml. portions of acetone, maintaining each portion in contact with the residue for approximately one minute with continuous stirring, before applying suction and until the following qualitative test for RDX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml. beaker, evaporate to dryness in a steam bath and add several drops of a solution of chromotropic acid in normal (N) sulfuric acid. The formation of a reddish-brown color as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying oven at 100° to 105° C. for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue to percentage of graphite on a moisture-free basis as follows:

$$\text{Percent graphite} = \frac{100A}{W}$$

where:

A = increase in weight of tared crucible.
W = weight of specimen on a moisture-free basis.

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4.4.3 Determination of Total Binder-Lubricant Content.-The percent total binder-lubricant content shall be determined by adding the percent calcium resinate or stearic acid obtained to the percent graphite, when applicable.

4.4.4 Determination RDX Content.-The percent RDX content shall be determined by subtracting the percent total binder-lubricant from 100 percent.

4.4.5 Determination of Moisture Content.-Transfer an accurately weighed portion of approximately 10.0 grams of the RDX composition from 4.3.3.3 to a tared weighing bottle with an outside ground cap. Place the bottle and contents with cap removed in an oven at 100 plus or minus 5 degrees C. for 40 to 50 minutes; cool in a desiccator, stopper and reweigh. Calculate the loss in weight of the bottle and contents as percent moisture in the sample.

4.4.6 Process functioning test.-The lead cup assembly shall be placed in the fixture shown on the Equipment List and a weight of 0.25 plus 0.01 ounce shall be dropped from a height of three (3) inches on the firing pin. Observation shall be made for satisfactory functioning evidenced by the size of the depression in the lead block (see 3.5).

4.4.7 Functioning.-The lead cup assembly shall be placed in the fixture shown on the Equipment List and a weight of 0.25 plus 0.01 ounce shall be dropped from a height of three (3) inches on the firing pin. Observation shall be made for satisfactory functioning evidenced by the size of the depression in the lead block (see 3.5).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level C.-The lead cup assemblies shall be preserved and packaged in accordance with the best commercial practice approved by the Contracting Officer and in accordance with Interstate Commerce Commission Regulation 49 CFR 171-178.

5.2 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

5.3 Marking.-Marking shall be in accordance with Interstate Commerce Commission Regulation 49 CFR 171-178.

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6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.- The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.- Equipment lists indicate the availability of inspection equipment designs by showing in the "Part number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an

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item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall reference the applicable five-digit code number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1C. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting

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officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Check test for deterioration.- If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the lead cup assemblies have been subjected to adverse conditions, however brief, at any time since previous tests, the lead cup assembly lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in Para. 4.4.7 immediately before the lead cup assembly lot is assembled into the fuze. This test shall be performed by the contractor on lead cup assemblies selected by the Government inspector at the facility assembling the lead cups into the fuze. (See 6.4.1). (Sampling Plan 500-14-15)

6.4.1 Cost of check test.- The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the lead cup assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the lead cup assemblies or to the contractor assembling the lead cup assemblies into the fuze, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc, permitted while the lead cup assembly lot was under the jurisdiction of either contractor.

6.5 Submission of results of contractor-conducted examinations and tests.- Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
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